

Assignment 1

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STAT4601/STAT5703 Assignment #1 Due Date: January 30, 2025 midnight

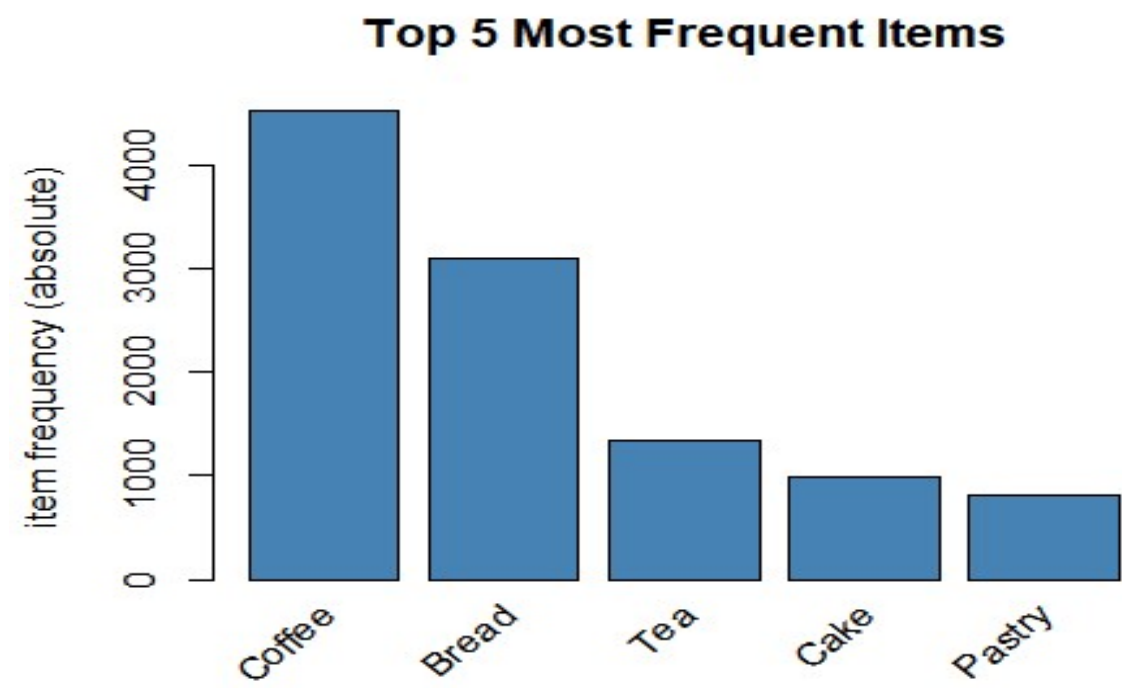
Introduction

This in-depth analysis delves into customer purchasing behavior within a bakery setting to extract actionable insights aimed at optimizing inventory management, enhancing product recommendations, and fine-tuning marketing strategies. Utilizing a robust dataset comprising 20,507 individual purchase records from over 9,000 transactions, the report leverages advanced analytical techniques to decode patterns and improve business operations.

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Frequent Item Analysis

	items	support	count
## [1]	{Coffee}	0.47818278	4526
## [2]	{Bread}	0.32688854	3094
## [3]	{Tea}	0.14252509	1349
## [4]	{Cake}	0.10385631	983
## [5]	{Bread, Coffee}	0.09001585	852
## [6]	{Pastry}	0.08600106	814
## [7]	{Sandwich}	0.07184363	680
## [8]	{Medialuna}	0.06170100	584
## [9]	{Hot chocolate}	0.05832013	552
## [10]	{Cake, Coffee}	0.05472795	518
## [11]	{Cookies}	0.05430534	514
## [12]	{Coffee, Tea}	0.04986793	472
## [13]	{Coffee, Pastry}	0.04743793	449



Top 5 Most Purchased Items (single item):

Coffee (best seller)

Bread (best seller)

Tea

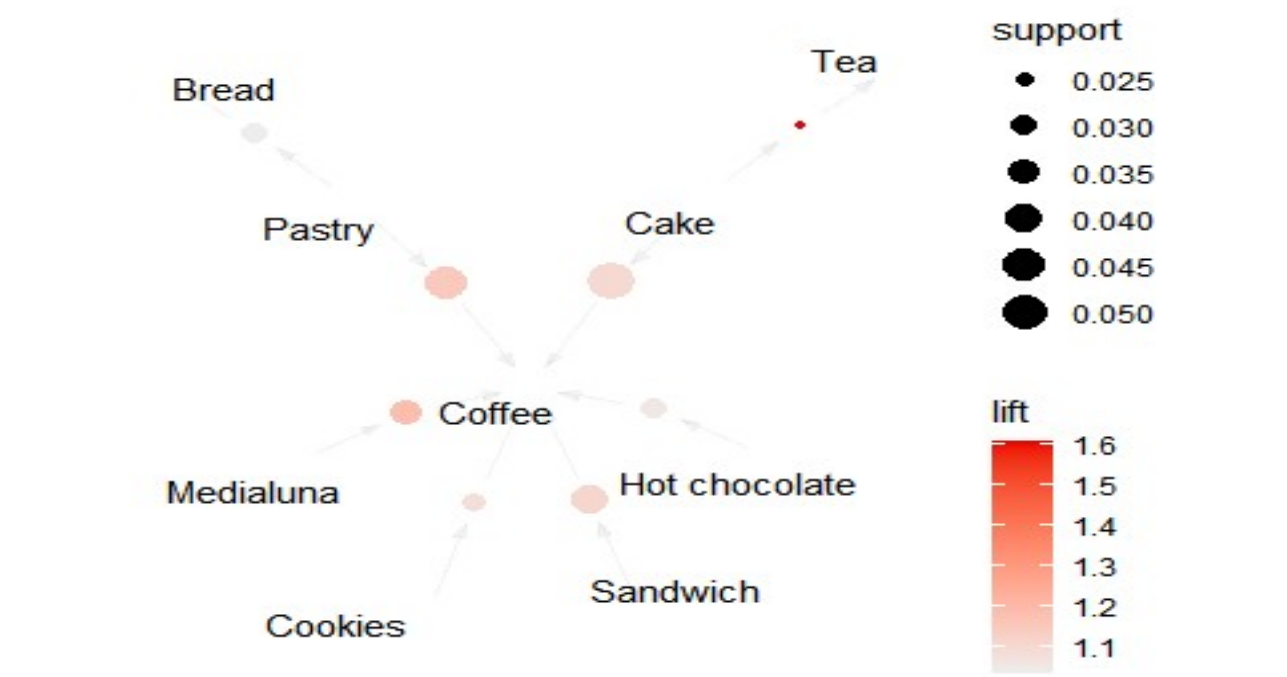
Cake

Pastry

Association Rule Mining

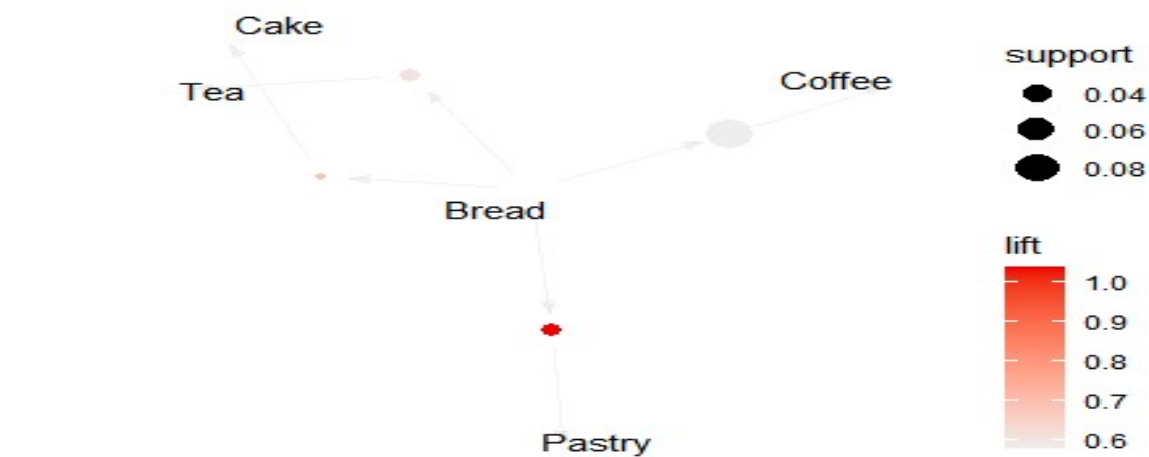
##	lhs	rhs	support	confidence	coverage	lift	count
## [1]	{Medialuna}	=> {Coffee}	0.03507660	0.5684932	0.06170100	1.188862	332
## [2]	{Pastry}	=> {Coffee}	0.04743793	0.5515971	0.08600106	1.153528	449
## [3]	{Sandwich}	=> {Coffee}	0.03824617	0.5323529	0.07184363	1.113283	362
## [4]	{Cake}	=> {Coffee}	0.05472795	0.5269583	0.10385631	1.102002	518
## [5]	{Cookies}	=> {Coffee}	0.02820919	0.5194553	0.05430534	1.086311	267
## [6]	{Hot chocolate}	=> {Coffee}	0.02958267	0.5072464	0.05832013	1.060779	280

## [7]	{Pastry}	=>	{Bread}	0.02916006	0.3390663	0.08600106	1.037254	27
6								
## [8]	{Cake}	=>	{Tea}	0.02377179	0.2288911	0.10385631	1.605971	22
5								



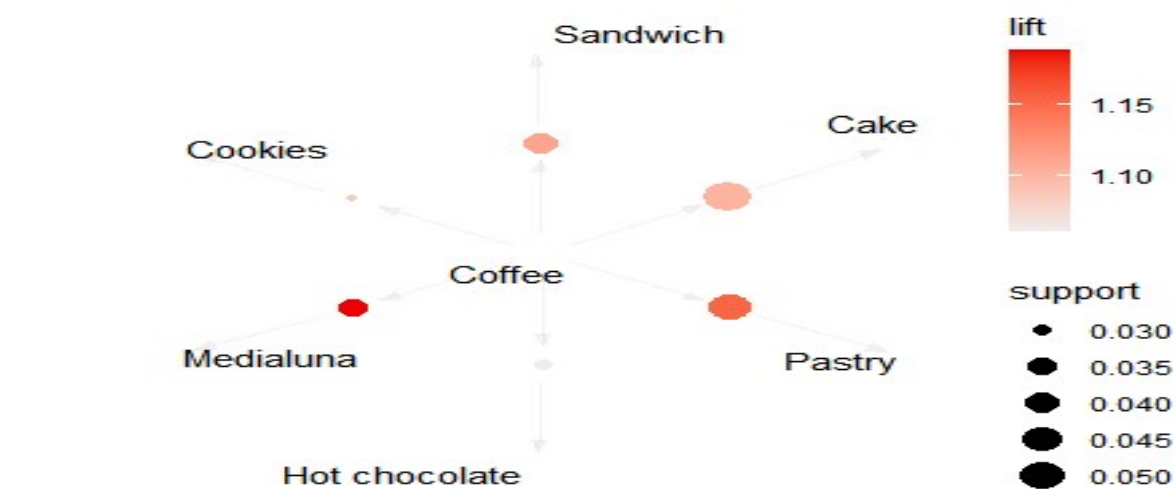
Guessing the future transaction if people buy bread or coffee (which are the most sold item)
for bread

##	lhs	rhs	support	confidence	coverage	lift	count
## [1]	{Bread}	=>	{Coffee}	0.09001585	0.27537169	0.3268885	0.5758712 852
## [2]	{Bread}	=>	{Pastry}	0.02916006	0.08920491	0.3268885	1.0372537 276
## [3]	{Bread}	=>	{Tea}	0.02810354	0.08597285	0.3268885	0.6032120 266
## [4]	{Bread}	=>	{Cake}	0.02334918	0.07142857	0.3268885	0.6877634 221



for coffee

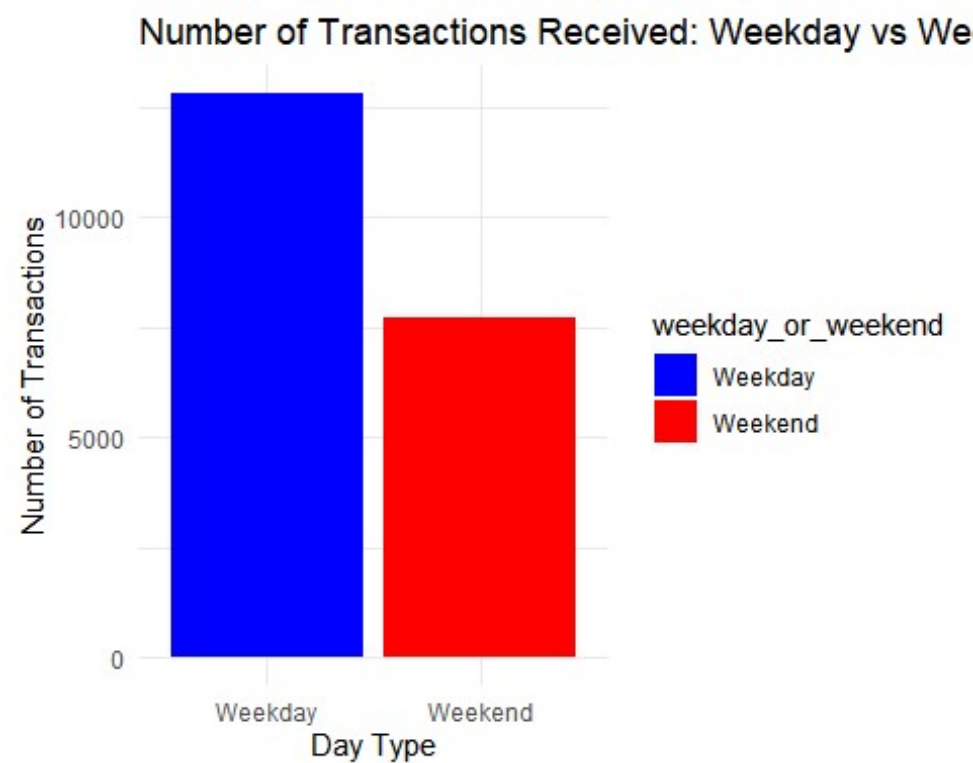
##	lhs	rhs	support	confidence	coverage	lift	count
## [1]	{Coffee} =>	{Cake}	0.05472795	0.11444985	0.4781828	1.102002	518
## [2]	{Coffee} =>	{Pastry}	0.04743793	0.09920460	0.4781828	1.153528	449
## [3]	{Coffee} =>	{Sandwich}	0.03824617	0.07998232	0.4781828	1.113283	362
## [4]	{Coffee} =>	{Medialuna}	0.03507660	0.07335395	0.4781828	1.188862	332
## [5]	{Coffee} =>	{Hot chocolate}	0.02958267	0.06186478	0.4781828	1.060779	280
## [6]	{Coffee} =>	{Cookies}	0.02820919	0.05899249	0.4781828	1.086311	267



Transaction Trends

Transactions by Weekday vs. Weekend

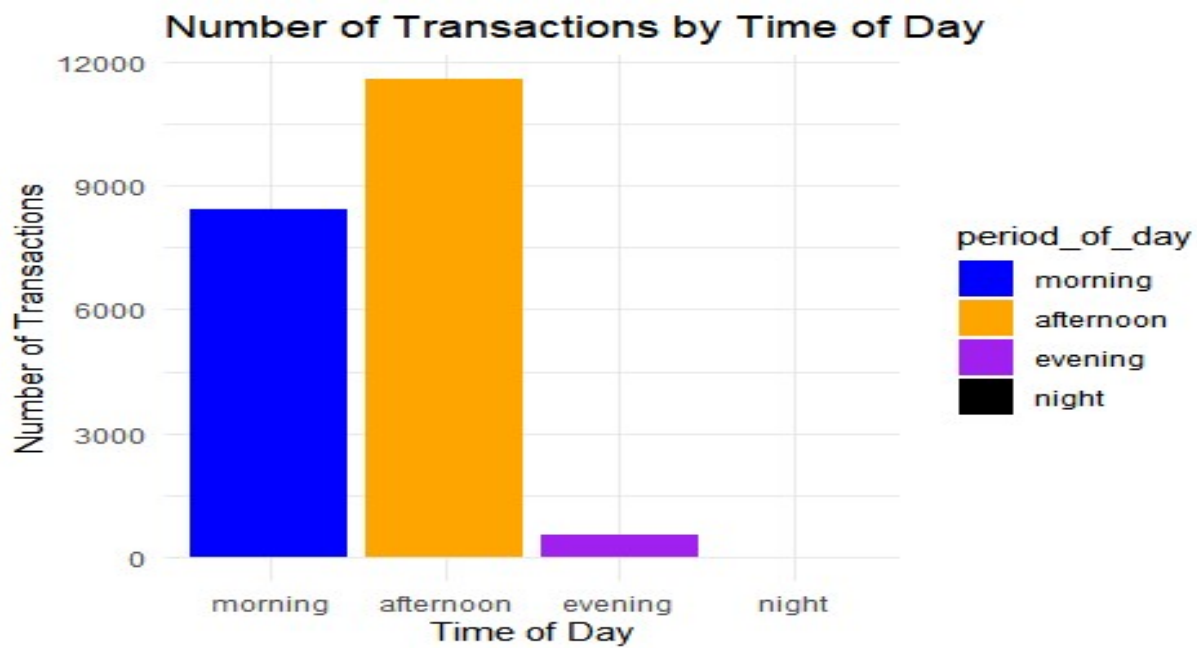
##	1 Weekday	12807
##	2 Weekend	7700



Transactions by Period of the Day

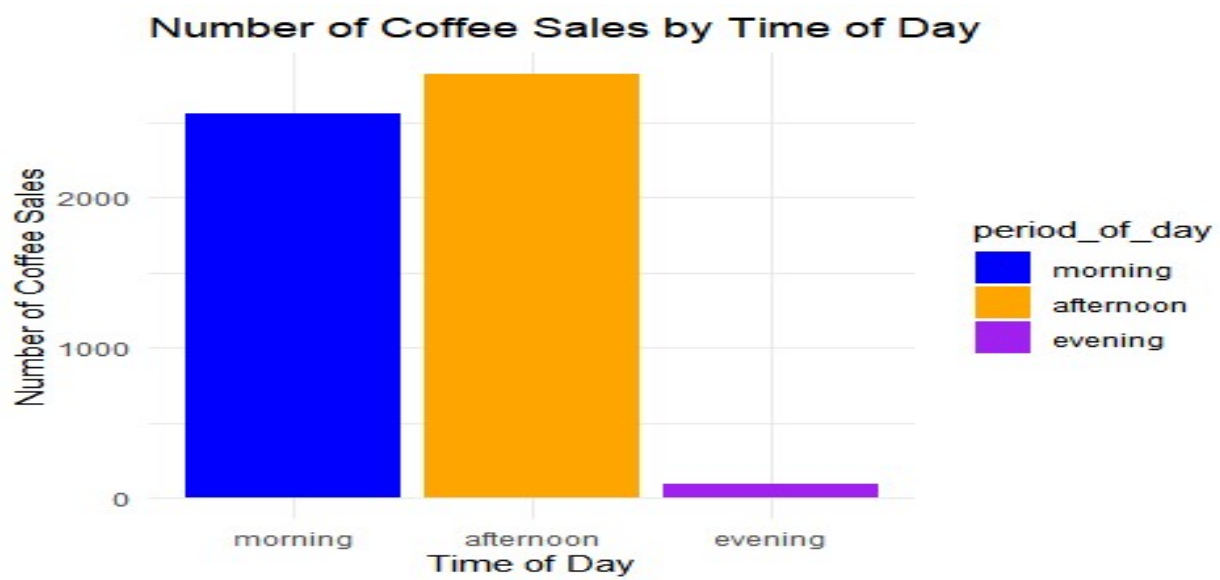
What times customer make the most purchase?

##	1 morning	8404
##	2 afternoon	11569
##	3 evening	520
##	4 night	14

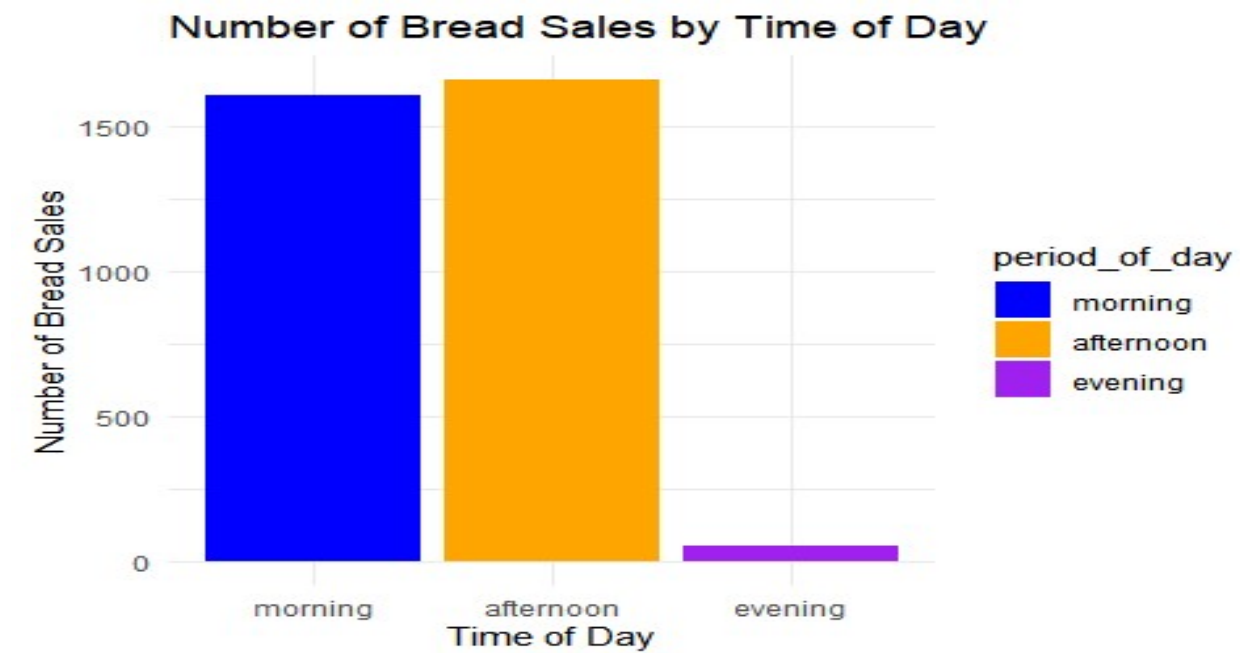


period of the day when each of the top 5 items is most frequently purchased.

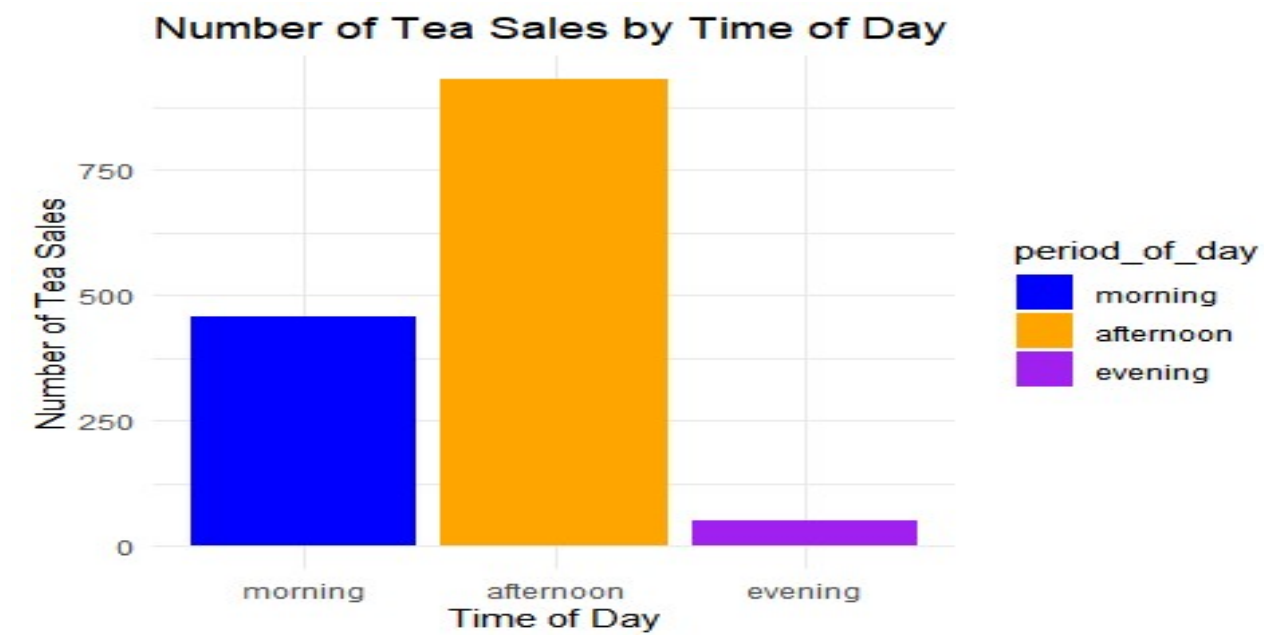
##	1	morning	2561
##	2	afternoon	2823
##	3	evening	87



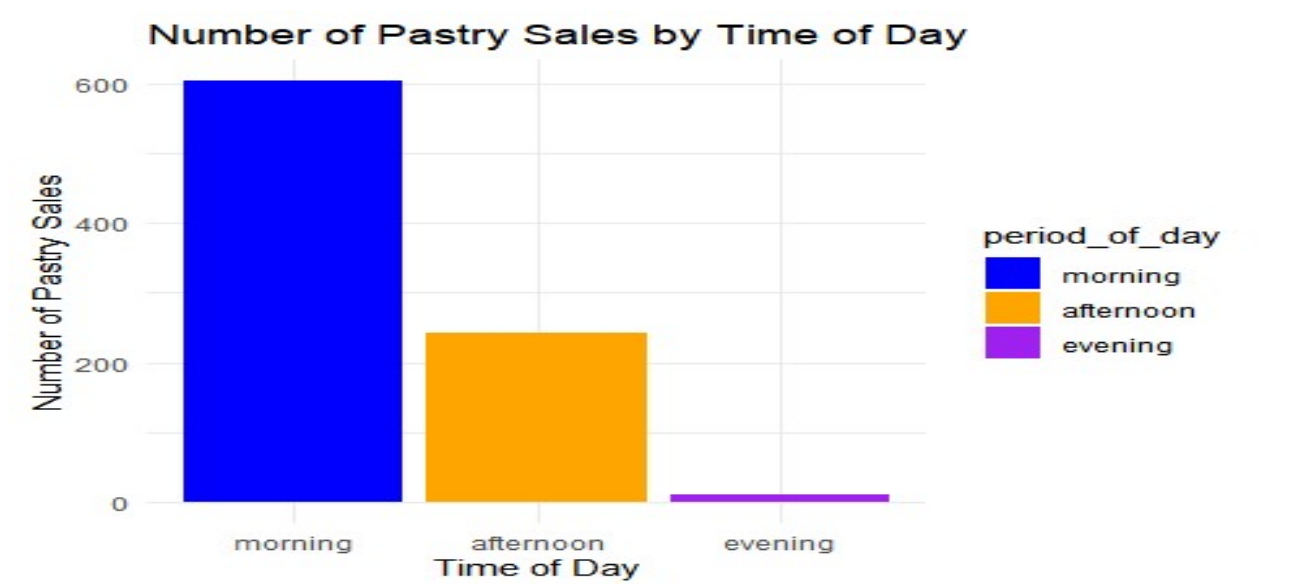
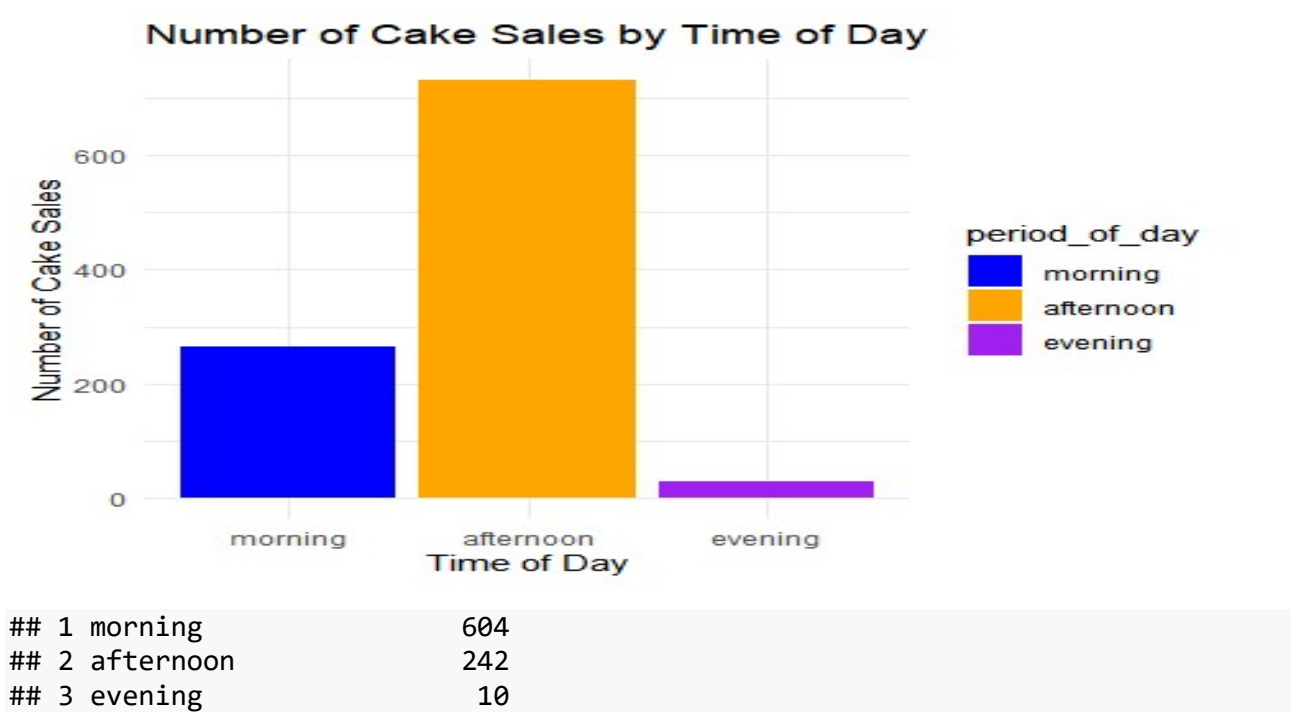
##	1	morning	1610
##	2	afternoon	1661
##	3	evening	54



##	1	morning	456
##	2	afternoon	930
##	3	evening	49



##	1	morning	264
##	2	afternoon	731
##	3	evening	30



Report

Market Basket Analysis Overview

Market basket analysis via association rule mining offers a rigorous approach to understanding customer buying patterns. This method is pivotal for:

- **Product Arrangement:** Strategically positioning related goods to maximize visibility and accessibility based on consumer purchasing habits.

- **Product Recommendations:** Enhancing the customer experience through personalized promotions based on reliable buying behavior data.
- **Inventory Management:** Employing predictive analytics to forecast purchasing trends and manage stock levels efficiently.
- **Advertising Optimization:** Refining marketing efforts to increase the efficacy of promotions and advertising, based on a deeper understanding of customer responses.

Key Metrics Explained

- **Support** quantifies how frequently items appear across all transactions, shedding light on the most popular products.
- **Confidence** measures the probability of purchasing item Y given that item X has been bought, which is crucial for understanding product pairings.
- **Lift** assesses the strength of an association between two products relative to their independent sales, helping to identify genuinely interconnected product purchases.

Detailed Findings

Frequent Item Analysis

- Coffee and Bread emerge as the most frequently purchased items, signaling their importance in daily sales. The visualization supports this, depicting Coffee as the predominant choice among patrons, followed by Bread, underscoring their foundational role in the bakery's offerings.

Association Rule Mining

- Key findings include:
 - Coffee's frequent combination with items like Medialuna, Pastry, and Cake suggests its central role in cross-selling strategies.
 - Bread commonly pairs with Pastry and Tea, indicating potential for bundling these items to increase sales.

Sales Trends

- Analysis of sales distribution indicates a higher transaction volume on weekdays with afternoon peaks, suggesting optimal times for targeted staffing and inventory adjustments.
- Morning sales are predominantly driven by Coffee, presenting an opportunity for targeted morning promotions to boost early day sales.

Strategic Insights and Business Recommendations

- **Product Placement and Merchandising:**
 - Strategic placement of complementary items like Coffee and Pastries near the entrance can stimulate impulse buys.
 - Creative displays that highlight popular combinations can enhance visual appeal and increase basket size.
- **Dynamic Promotional Strategies:**
 - Time-specific promotions, especially during peak sales periods, can maximize sales potential.
 - Bundling top-selling items during peak times can also encourage larger purchases.
- **Inventory and Stock Management:**
 - Synchronize inventory replenishment with peak sales times to ensure availability, especially for best-sellers like Coffee and Bread.
 - Leverage insights from association rule mining to forecast demand and manage stock levels proactively.
- **Enhanced Customer Engagement:**
 - A loyalty program that rewards frequent purchases could foster customer retention and increase overall sales.
 - Data-driven marketing campaigns can be tailored to customer preferences, maximizing engagement and boosting sales.

Conclusion

The exploration of customer purchasing behavior at the bakery has provided profound insights that are instrumental in steering strategic decisions across product placement, marketing, and inventory management. By synchronizing business strategies with demonstrated customer preferences and behaviors, the bakery is well-equipped to enhance operational efficiency, improve customer satisfaction, and drive sales growth. These adjustments are expected to not only boost the bakery's profitability but also enhance its competitive edge in the market. This analysis serves as a cornerstone for ongoing improvements and underscores the importance of data-driven decision making in modern retail environments.